

# House Price Prediction Analysis

From the correlation analysis and feature importance chart we can infer that "area" was the most influential factor affecting house prices . Bathrooms and Airconditioning also contributed significantly to price prediction.

The linear Regression model achieved a  $R^2$  score of about 0.65, meaning it was able to explain around 65% of the variations in house prices. The Random Forest model achieved a score of 0.61 which really surprised me

I initially expected Random Forest to produce better results as there are many factors influencing house price and I thought Linear Regression would fail and Random forest would produce much accurate results, but actually my predictions were wrong; linear regression performed better.

Based on these findings, a real estate business should focus on highlighting property area and key amenities when pricing or marketing houses, as these factors appear to have the strongest impact on property value.